R

REFRACTIVE SURGERY

StandardEyes
Refractive surgery analysis

User manual & walkthrough

By Noyman DBE, Mimouni M, Safir M

What this app does

One tool from raw Excel to publication-ready tables and figures

1 **Compare groups**

Summary tables for any number of groups, with the right test chosen per variable type.

2 **Match fairly**

Propensity-score matching (MatchIt) with a live matched-count preview and balance reporting.

3 **Model properly**

Clustered GEE so both eyes of a patient are handled correctly, plus covariate-adjusted regression.

4 **Publish**

The StandardEyes figures — every group on one chart, each panel separately and combined, JCRS double-angle astigmatism plots, semicircle vector plots and stability over time.

Before you start

Almost everything is already inside the folder

You need R 4.5.x

The statistics run in R, so R must be installed on your computer.

Use R 4.5 (not 4.6 or later):
cran.r-project.org/bin/windows/base/old/4.5.2/

Typical location:
C:\Program Files\R\R-4.5.2\

The app finds it automatically. If it can't, it asks you — see the next slide.

Already included

- The Python runtime that serves the app
- All 48 R packages the analysis needs
- The full statistical engine and figure code

Nothing is installed on your machine and nothing is downloaded — it works on a restricted network.

Starting the app

Three steps, once

1

Copy the whole folder

Put it in a folder you can write to, e.g. C:\StandardEyes or Documents — not C:\Program Files. Keep every subfolder; don't run it from inside a ZIP.

2

Double-click the launcher

Start StandardEyes.bat

3

Work in your browser

It opens automatically on a free port, shown in the console window — so it never clashes with another app.

Keep the black console window open while you work. Closing it stops the app.

If the app can't find R

A one-time Setup screen appears — you can also reopen it any time

You have three ways to point it at R:

Pick from the list	Any R installation found on the computer is listed — click one.
Browse…	Opens a file picker. Choose Rscript.exe inside R's bin folder.
Paste the path	Type it in and press Use this.

The app checks the choice immediately and reports the R version it found.

No R at all? Install R 4.5.x for Windows, then click Re-check. No extra R packages are ever needed — this app brings its own.

Reopen later with the ⚙ R setup button at the top right.

The workflow

Eight tabs, left to right — top to bottom here

1

Data & Groups

Load Excel, define groups

2

Variables

Baseline / Preop / Outcome

3

Filters

Inclusion rules + subsets

4

Matching

Propensity-score matching

5

Regression

Adjusted GEE

6

GEE

Clustered comparison

7

Special outputs

Figures

8

Run & Results

Run, preview, save

Steps 1–2 · Data and variables

1 Data & Groups

Load your Excel file (it stays on your computer).

Choose the column that defines the groups, then map its values into the groups you want to compare.

Set the reference group — every comparison is made against it.

Also set the patient ID, so both eyes of one patient are clustered correctly.

2 Variables & Types

Put each variable into Baseline, Preop or Outcome, and give it a type:

continuous · ordinal · dichotomous · categorical

The type decides which test is used and how the value is displayed, so this is the step that shapes your tables.

Also say how your file writes visual acuity: decimal, logMAR, Snellen or ETDRS letters.

Steps 3–4 · Filters and matching

3 Filters & Subsets

Add inclusion rules — age, follow-up, first treatment, cylinder limits, and so on.

Row counts update live, per group, so you can see exactly how many eyes each rule removes before running anything.

Subsets (e.g. PRK and LASIK) produce a separate full analysis each.

4 Matching

Matching against the reference group: nearest neighbour, exact, coarsened exact or subclassification.

Choose the covariates to match on and, for nearest neighbour, the ratio and the caliper. A live preview shows how many eyes match before you commit.

Balance is reported so you can show the groups became comparable.

Steps 5–6 · Regression and GEE

How to handle groups that still differ at baseline

GEE is how the regression is fitted, not an alternative to it. The model is outcome ~ group + covariates; GEE additionally accounts for both eyes coming from the same patient.

- **Adjust for what is unbalanced** — Put the baseline variables that still differ into the regression covariates. If the group effect survives, it is not explained by those confounders.
- **Match first, then adjust** — Regression adjustment after matching: adjust the outcome model for whatever matching left imbalanced — the strongest option the app offers.
- **Choose covariates by balance, not by p** — Add a covariate because it is still imbalanced (SMD above ~0.1), never because it changes the result.

Step 7 · Figures

One coherent suite — every option independently switchable

The composite

EVERY group on one chart, however many. Omnibus + versus-reference statistics, clustered by patient. Panel sets: Full (nine), Refractive (six), Cataract (four); stability joins as an extra panel.

Double-angle plots

The JCRS-2025 astigmatism standard: centroid, total SD, 95% nonparametric convex polygon and a dashed 95% CI of the centroid, per group.

Semicircle plots

Single-angle polar vector diagrams — TIA, SIA, difference vector and correction index — groups superimposed on a grey background, kept for familiarity.

Export tabs

Optional worksheets in mEYEstro and AstigMATIC format, for verifying numbers against those tools.

Stability over time is mapped once and joins the composite as its own panel. Every panel is saved as its own TIFF (300 dpi default) and again inside the composite.

Step 8 · Run, preview, save

Saved only when you say so

Results are held inside the app first, so you can look through every sheet before anything reaches your output folder.

Meanwhile a temporary copy sits in the app's own folder; it is deleted at your next run or when you close the app.

When you are happy, choose the output folder and save.

Progress and a full log are shown while R runs.

What you get

- Excel workbooks — summary, matching, GEE, regression
- A matched-only workbook with every pairing
- A figure folder per run, previewed in the app
- The filtered dataset as CSV (optional)

Adjusted effects and p-values are embedded in the main outcome table.

Presets — one per study

Save every setting of a project and reload it in one click

A preset stores everything: groups, variable typing, filters, subsets, matching, regression and figure choices.

- **Save** — Name the study and click save. It is stored inside the app on your own computer.
- **Reload** — Click the preset to restore the entire configuration.
- **Share** — Download the config as JSON and send it to a colleague — they load it and get your exact setup.

Seven worked ‘Example – …’ presets are included (built on the bundled example dataset); presets you save appear beside them.

If something goes wrong

“Python runtime is missing”	The folder was copied incompletely, or is still inside a ZIP. Copy the whole folder again.
“Cannot write to its own folder”	Move the whole folder somewhere you can write to (Documents, C:\StandardEyes) — not C:\Program Files.
The browser shows nothing	Wait a few seconds, then reload the address printed in the console window.
“R is not set up yet”	Click ⚙ R setup at the top right and point it at Rscript.exe.
A red “out of date” bar	Close the console window, start the app again, then press Ctrl+F5.
The analysis fails immediately	Check your R version is 4.5.x and the VA notation of your file — the log says why.
Am I in the right app?	The header says StandardEyes — By Noyman DBE, Mimouni M, Safir M — and a version such as v1.0.0 (build 2026-10-02-se.9).

Everything runs on your own computer.

No data leaves the machine, no internet connection is needed, and no packages are ever installed.

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